

Math 6601 Homework 2

Jacob Hauck

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Problem 1.

(a) Let $f(x) = -x^3 - \cos(x)$, and let $x_0 = -1$. Newton's method for solving $f(x) = 0$ is given by

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = x_k + \frac{x_k^3 + \cos(x_k)}{-3x_k^2 + \sin(x_k)}, \quad k \geq 0.$$

Then

$$x_1 = -1 + \frac{-1 + \cos(-1)}{-3 + \sin(-1)} = -\frac{2 + \sin(1) + \cos(1)}{3 + \sin(1)},$$

and

$$x_2 = -\frac{2 + \sin(1) + \cos(1)}{3 + \sin(1)} + \frac{\left(-\frac{2 + \sin(1) + \cos(1)}{3 + \sin(1)}\right)^3 + \cos\left(-\frac{2 + \sin(1) + \cos(1)}{3 + \sin(1)}\right)}{-3\left(-\frac{2 + \sin(1) + \cos(1)}{3 + \sin(1)}\right)^2 + \sin\left(-\frac{2 + \sin(1) + \cos(1)}{3 + \sin(1)}\right)} \approx -0.865684.$$

Newton's method would not work with $x_0 = 0$ because $f'(x_0) = 0$.

(b) Let

$$f(x) = \int_0^x e^{t^2} dt - 1.$$

Then Newton's method can be used to solve for $f(x) = 0$, or to find x such that $\int_0^x e^{t^2} dt = 1$. Given an initial guess x_0 , the iteration is given by

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = x_k - \frac{\int_0^{x_k} e^{t^2} dt - 1}{e^{x_k^2}}, \quad k \geq 0$$

because $f'(x) = e^{x^2}$ by the fundamental theorem of calculus.

Problem 2.

If f is $m + 1$ times continuously differentiable on $[a, b]$, which contains a root z of f of multiplicity $m > 1$, then Modified Newton's Method

$$x_{k+1} = x_k - \frac{mf(x_k)}{f'(x_k)}, \quad k \geq 0$$

converges locally and quadratically to z .

Proof. Using the Taylor expansion of f about z and of f' about z and the fact that z has multiplicity m , we have

$$\begin{aligned} f(x_k) &= f(z) + f'(z)(x_k - z) + \cdots + \frac{1}{m!}f^{(m)}(z)(x_k - z)^m + \frac{1}{(m+1)!}f^{(m+1)}(\xi_1(z))(x_k - z)^{m+1} \\ &= \frac{1}{m!}f^{(m)}(z)(x_k - z)^m + \frac{1}{(m+1)!}f^{(m+1)}(\xi_1(x_k))(x_k - z)^{m+1} \end{aligned}$$

for some $\xi_1(x_k)$ such that $|\xi_1(x_k) - z| \leq |x_k - z|$, and

$$\begin{aligned} f'(x_k) &= f'(z) + f''(z)(x_k - z) + \cdots + \frac{1}{(m-1)!}f^{(m)}(z)(x_k - z)^{m-1} + \frac{1}{m!}f^{(m+1)}(\xi_2(z))(x_k - z)^m \\ &= \frac{1}{(m-1)!}f^{(m)}(z)(x_k - z)^{m-1} + \frac{1}{m!}f^{(m+1)}(\xi_2(x_k))(x_k - z)^m \end{aligned}$$

for some $\xi_2(x_k)$ such that $|\xi_2(x_k) - z| \leq |x_k - z|$. Then

$$\begin{aligned} |x_{k+1} - z| &= \left| x_k - z - m \frac{f(x_k)}{f'(x_k)} \right| \\ &= \left| x_k - z - m \frac{\frac{1}{m!}f^{(m)}(z)(x_k - z)^m + \frac{1}{(m+1)!}f^{(m+1)}(\xi_1(x_k))(x_k - z)^{m+1}}{\frac{1}{(m-1)!}f^{(m)}(z)(x_k - z)^{m-1} + \frac{1}{m!}f^{(m+1)}(\xi_2(z))(x_k - z)^m} \right| \\ &= \left| x_k - z - \frac{(m+1)f^{(m)}(z)(x_k - z) + f^{(m+1)}(\xi_1(x_k))(x_k - z)^2}{(m+1)f^{(m)}(z) + \frac{m+1}{m}f^{(m+1)}(\xi_2(z))(x_k - z)} \right| \\ &= \left| \frac{\frac{m+1}{m}f^{(m+1)}(\xi_2(z))(x_k - z)^2 - f^{(m+1)}(\xi_1(x_k))(x_k - z)^2}{(m+1)f^{(m)}(z) + \frac{m+1}{m}f^{(m+1)}(\xi_2(z))(x_k - z)} \right| \\ &\leq \left| (m+1)f^{(m)}(z) + \frac{m+1}{m}f^{(m+1)}(\xi_2(z))(x_k - z) \right|^{-1} \left(\frac{m+1}{m} \left| f^{(m+1)}(\xi_2(z)) \right| + \left| f^{(m+1)}(\xi_1(x_k)) \right| \right) |x_k - z|^2. \end{aligned}$$

Since $f^{(m+1)}$ is continuous on $[a, b]$ it is bounded, say by $C_{m+1} > 0$. Thus, given $\delta > 0$, then $|x_k - z| < \delta$ implies that

$$\left| (m+1)f^{(m)}(z) + \frac{m+1}{m}f^{(m+1)}(\xi_2(x_k))(x_k - z) \right| \geq (m+1) \left| f^{(m)}(z) \right| - \frac{m+1}{m}C_{m+1}\delta.$$

Since $f^{(m)}(z) \neq 0$ (because the multiplicity of z is m), we can choose $\delta > 0$ small enough that the quantity on the right is positive. In this case, $|x_k - z| < \delta$ implies that

$$\left| (m+1)f^{(m)}(z) + \frac{m+1}{m}f^{(m+1)}(\xi_2(x_k))(x_k - z) \right|^{-1} \leq \frac{1}{(m+1) \left| f^{(m)}(z) \right| - \frac{m+1}{m}C_{m+1}\delta} =: D.$$

Then setting

$$C = DC_{m+1} \left(\frac{m+1}{m} + 1 \right),$$

we obtain

$$|x_{k+1} - z| \leq C|x_k - z|^2$$

if $|x_k - z| < \delta$. Thus, we can choose $\delta' > 0$ such that if $x_0 \in (z - \delta', z + \delta') \cap [a, b]$, then $x_k \rightarrow z$ quadratically. In particular, choose $\delta' > 0$ such that $\delta' < \min\{\delta, \frac{1}{C}\}$. $\square$

Problem 3.

Let

$$f(x) = \frac{1}{2} + \frac{1}{4}x^2 - x \sin(x) - \frac{1}{2} \cos(2x).$$

- (a) Running Newton's method using the code in `problem3a_output.txt`, it takes 18 steps to reach an error $|x_{k+1} - x_k| < 10^{-6}$. Steps are presented in Table 1

k	x_k
0	1.57079632679490
1	1.78539816339745
2	1.84456162960179
3	1.87083441768886
4	1.88334642692721
5	1.88946376418230
6	1.89248962453426
7	1.89399456835593
8	1.89474506969115
9	1.89511983090939
10	1.89530708955216
11	1.89540068843122
12	1.89544748026546
13	1.89547087428599
14	1.89548257081704
15	1.89548841897691
16	1.89549134299450
17	1.89549280504346
18	1.89549353608069

Table 1: Steps for Newton's method for solving $f(x) = 0$ with $x_0 = \frac{\pi}{2}$

- (b) The convergence seems to be too slow for Newton's method. Indeed, by solving to a much lower error (10^{-12}) to get a good approximation of the true root z , we can estimate the error on each step above and compute the convergence rate. This is presented in Table 2 (the code is also in `problem3a_output.txt`). The result suggests that Newton's method is converging linearly with rate $\frac{1}{2}$.

This is corroborated by the fact that $f'(z) = 0$; in fact, by the double-angle identity,

$$f(x) = \frac{1}{4}x^2 - x \sin(x) + \sin^2(x) = \left(\frac{1}{2}x - \sin(x)\right)^2,$$

so $f(z) = 0$ implies that $\frac{1}{2}z - \sin(z) = 0$. Then

$$f'(x) = 2 \left(\frac{1}{2}x - \sin(x)\right) \left(\frac{1}{2} - \cos(x)\right) \implies f'(z) = 0.$$

Thus, Newton's method should converge linearly, as we observed.

- (c) If we can determine the multiplicity m of the root z , then we can use improved Newton's method from Problem 2 to achieve quadratic convergence again. We already know that $m \geq 2$. Now, we have

$$f''(x) = 2 \left(\frac{1}{2} - \cos(x)\right)^2 + 2 \left(\frac{1}{2}x - \sin(x)\right) \sin(x).$$

k	$\frac{ x_{k+1}-z }{ x_k-z }$
0	0.339072360071223
1	0.462619750185422
2	0.484165857812748
3	0.492615943053683
4	0.496425510330024
5	0.498239941979161
6	0.499125462815884
7	0.499561850963095
8	0.499776240554527
9	0.499878023019502
10	0.499918634151070
11	0.499918522333072
12	0.499877563762001
13	0.499775562212887
14	0.499559963878091
15	0.499129833394247
16	0.498242847250807
17	0.496464515761668

Table 2: Estimated step-by-step convergence rate for Newton's method from **(a)**

k	x_k
0	1.57079632679490
1	2.00000000000000
2	1.90099559420391
3	1.89551164537963
4	1.89549426721330
5	1.89549380571069

Table 3: Modified Newton's method steps for solving $f(x) = 0$.

Thus, $\frac{1}{2}z - \sin(z) = 0$ implies that

$$f''(z) = 2 \left(\frac{1}{2} - \cos(z) \right)^2.$$

If $\cos(z) = \frac{1}{2}$, then $z = \pm\frac{\pi}{3} + 2\pi n$, where n is an integer. Then $\sin(z) = \pm\frac{\sqrt{3}}{2} \neq \frac{1}{2} (\pm\frac{\pi}{3} + 2\pi n)$ for any choice of signs or value of n , meaning that the condition $\frac{1}{2}z - \sin(z) = 0$ is violated. Thus, we cannot have $\cos(z) = \frac{1}{2}$, so $f''(z) \neq 0$. This means that $m = 2$, and we can use modified Newton's method from Problem 2 to achieve enhanced convergence.

This is implemented in `problem3c_output.txt`. Numerical results are also presented in Table 3. As expected, it took far fewer steps to achieve a tolerance of 10^{-6} (only 5 steps).

Problem 4.

Let $f(x) = (x - \sin(x))(x - \cos(x))$.

- (a) Running Newton's method (see `problem4a_output.txt`) with $x_0 = 0.6$ exhibits quadratic convergence, as can be seen in the estimated convergence orders in Table 4.

k	Approximate order
1	0.51790088486251
2	1.67903770973958
3	1.56906120587407
4	1.58561581327327
5	1.68105326786349
6	1.80036881025641
7	1.88878749395772

Table 4: Estimated convergence order for Newton's method for solving $f(x) = 0$ with $x_0 = 0.6$

- (b) Running Newton's method (see `problem4b_output.txt`) with $x_0 = 0.1$ seems to exhibit linear convergence. This is likely because it is converging to the root $z = 0$, and $f'(0) = 0$ because

$$f'(x) = (1 - \cos(x))(x - \cos(x)) + (x - \sin(x))(1 + \sin(x)) \implies f'(0) = (1 - 1)(0 - 1) + (0 - 0)(1 + 0) = 0,$$

which implies that Newton's method will converge linearly if x_0 is near 0. Noting that

$$\begin{aligned} f''(x) &= 1 + \sin(x) + \sin(x)(x - \cos(x)) - \cos(x)(1 + \sin(x)) + 1 - \cos(x) + \cos(x)(x - \sin(x)) + \sin(x)(1 - \cos(x)) \\ &= 2 + 2(\sin(x) - \cos(x) - \sin(x)\cos(x)) + \sin(x)(x - \cos(x)) + \cos(x)(x - \sin(x)), \end{aligned}$$

it follows that

$$f''(0) = 0.$$

Furthermore,

$$f'''(x) = 2(\cos(x) + \sin(x) - \cos(2x)) + \cos(x)(x - \cos(x)) + \sin(x)(1 + \sin(x)) - \sin(x)(x - \sin(x)) + \cos(x)(1 - \cos(x)),$$

so

$$f'''(0) = 2(1 + 0 - 1) + 1(0 - 1) + 0(1 + 0) - 0(0 - 0) + 1(1 - 1) = -1 \neq 0.$$

Then 0 is a root of multiplicity $m = 3$, and we can use modified Newton's method with $m = 3$ to obtain quadratic convergence. This can be seen in `problem4b_output.txt`, in which we see that the modified method requires 4 steps to reach the same tolerance as the original Newton's method did in 36 steps.

Problem 5.

Let α be a root of f , and suppose that $f'(\alpha) \neq 0$ and $f''(\alpha) \neq 0$, and f is three times continuously differentiable on an interval $[a, b]$ containing α . Define the iteration

$$x_{k+1} = z_{k+1} - \frac{f(z_{k+1})}{f'(x_k)}, \quad z_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

Then $x_k \rightarrow \alpha$ locally with order 3.

Proof. By Taylor's theorem, we have for $x \in U$,

$$f(x) = f'(\alpha)(x - \alpha) + \frac{1}{2}f''(\alpha)(x - \alpha)^2 + \frac{1}{6}f'''(\xi_1(x))(x - \alpha)^3$$

for some $\xi_1(x) \in U$ satisfying $|\xi_1(x) - \alpha| \leq |x - \alpha|$, and

$$f'(x) = f'(\alpha) + f''(\alpha)(x - \alpha) + \frac{1}{2}f'''(\xi_2(x))(x - \alpha)^2$$

for some $\xi_2(x) \in U$ satisfying $|\xi_2(x) - \alpha| \leq |x - \alpha|$. Then

$$\begin{aligned} z_{k+1} &= \alpha + x_k - \alpha - \frac{f'(\alpha)(x_k - \alpha) + \frac{1}{2}f''(\alpha)(x_k - \alpha)^2 + \frac{1}{6}f'''(\xi_1(x_k))(x_k - \alpha)^3}{f'(\alpha) + f''(\alpha)(x_k - \alpha) + \frac{1}{2}f'''(\xi_2(x_k))(x_k - \alpha)^2} \\ &= \alpha + \frac{\frac{1}{2}f''(\alpha) + \left(\frac{1}{2}f'''(\xi_2(x_k)) + \frac{1}{6}f'''(\xi_1(x_k))\right)(x_k - \alpha)}{f'(\alpha) + f''(\alpha)(x_k - \alpha) + \frac{1}{2}f'''(\xi_2(x_k))(x_k - \alpha)^2}(x_k - \alpha)^2 \\ &= \alpha + c_1(x_k)(x_k - \alpha)^2, \end{aligned}$$

where

$$c_1(x) = \frac{\frac{1}{2}f''(\alpha) + \left(\frac{1}{2}f'''(\xi_2(x)) + \frac{1}{6}f'''(\xi_1(x))\right)(x - \alpha)}{f'(\alpha) + f''(\alpha)(x - \alpha) + \frac{1}{2}f'''(\xi_2(x))(x - \alpha)^2}.$$

Then substituting for z_{k+1} , we obtain

$$\begin{aligned} x_{k+1} - \alpha &= c_1(x_k)(x_k - \alpha)^2 - \frac{f(\alpha + c_1(x_k)(x_k - \alpha)^2)}{f'(x_k)} \\ &= c_1(x_k)(x_k - \alpha)^2 - \frac{f'(\alpha)c_1(x_k)(x_k - \alpha)^2 + \frac{1}{2}f''(\alpha)c_1^2(x_k)(x_k - \alpha)^4 + \frac{1}{6}f'''(\xi_1(x_k))c_1^3(x_k)(x_k - \alpha)^6}{f'(\alpha) + f''(\alpha)(x_k - \alpha) + \frac{1}{2}f'''(\xi_2(x_k))(x_k - \alpha)^2} \\ &= \frac{f''(\alpha)c_1(x_k) + \left(\frac{1}{2}f'''(\xi_2(x_k))c_1(x_k) - \frac{1}{2}f''(\alpha)c_1^2(x_k)\right)(x_k - \alpha) - \frac{1}{6}f'''(\xi_1(x_k))c_1^3(x_k)(x_k - \alpha)^3}{f'(x_k)}(x_k - \alpha)^3. \end{aligned}$$

Now, f' is continuous on U , so $f'(x) \rightarrow f'(\alpha) \neq 0$ as $x \rightarrow \alpha$. Moreover, since f''' is continuous on $[a, b]$, it is bounded. This implies that c_1 is continuous at α , and clearly $c_1(x) \rightarrow \frac{f''(\alpha)}{2f'(\alpha)}$ as $x \rightarrow \alpha$. Since the entire coefficient of $(x_k - \alpha)^3$ above depends continuously on x_k and $c_1(x_k)$, it follows that the coefficient is continuous at α . Then there exists some constant $C > 0$ and $\delta > 0$ so that $|x_k - \alpha| < \delta$ implies

$$|x_{k+1} - \alpha| \leq C|x_k - \alpha|^3.$$

Then $x_k \rightarrow \alpha$ locally and with order 3. □

Problem 6.

Let $F(x) = (x_1^2 + x_2^2 - 2, -x_1 + x_2)^T$.

(a) First, note that

$$DF(x) = \begin{bmatrix} 2x_1 & 2x_2 \\ -1 & 1 \end{bmatrix},$$

and

$$DF^{-1}(x) = \frac{1}{2(x_1 + x_2)} \begin{bmatrix} 1 & -2x_2 \\ 1 & 2x_1 \end{bmatrix}.$$

If the initial guess is $x^{(0)} = (2, 2)^T$, then the first step of Newton's method is

$$x^{(1)} = x^{(0)} - DF^{-1}(x^{(0)}) F(x^{(0)}) = \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \frac{1}{2(2+2)} \begin{bmatrix} 1 & -4 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{5}{4} \\ \frac{5}{4} \end{bmatrix}.$$

The next step is then

$$x^{(2)} = x^{(1)} - DF^{-1}(x^{(1)}) F(x^{(1)}) = \begin{bmatrix} \frac{5}{4} \\ \frac{5}{4} \end{bmatrix} - \frac{1}{2(\frac{5}{4} + \frac{5}{4})} \begin{bmatrix} 1 & -\frac{10}{4} \\ 1 & \frac{10}{4} \end{bmatrix} \begin{bmatrix} \frac{9}{8} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{41}{40} \\ \frac{41}{40} \end{bmatrix}.$$

(b) Based on the output in `problem6b_output.txt`, it appears that Newton's method is converging quadratically.

(c) If $F(x) = 0$, then $x_1 = x_2$ by the second component being 0, and then $x_1^2 = 1$ by the first component being zero, leading to two solutions

$$z^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad z^{(2)} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

(d) We have $\det(DF(z)) = 2(x_1 + x_2)$, so $\det(DF(z^{(1)})) = 4 \neq 0$, and $\det(DF(z^{(2)})) = -4 \neq 0$.

Problem 7.

Let $F(x) = (x_1^2 + 2x_2^2, -4x_1 + x_2)$.

(a) First, note that

$$DF(x) = \begin{bmatrix} 2x_1 & 4x_2 \\ -4 & 1 \end{bmatrix},$$

and

$$DF^{-1}(x) = \frac{1}{2x_1 + 16x_2} \begin{bmatrix} 1 & -4x_2 \\ 4 & 2x_1 \end{bmatrix}.$$

Then with initial guess $x^{(0)} = (2, 2)^T$, we have

$$x^{(1)} = x^{(0)} - DF^{-1}(x^{(0)}) F(x^{(0)}) = \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \frac{1}{36} \begin{bmatrix} 1 & -8 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} 12 \\ -6 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{4}{3} \end{bmatrix}.$$

Then the next step is

$$x^{(2)} = x^{(1)} - DF^{-1}(x^{(1)}) F(x^{(1)}) = \begin{bmatrix} \frac{1}{3} \\ \frac{4}{3} \end{bmatrix} - \frac{1}{22} \begin{bmatrix} 1 & -\frac{4}{3} \\ 4 & \frac{8}{3} \end{bmatrix} \begin{bmatrix} \frac{11}{3} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{6} \\ \frac{2}{3} \end{bmatrix}.$$

(b) Based on the numerical results in `problem7b_output.txt`, it seems that the convergence is linear.

(c) The first component of $F(x)$ can only be zero if $x_1 = x_2 = 0$, which makes the second component zero. Thus, $z = (0, 0)^T$ is the only root of F .

(d) We have $\det(DF(z)) = 2z_1 + 16z_2 = 0$.

(e) Based on the calculations in `problem7e_output.txt`, it seems that the convergence rate is 0.5.